The Uniform Walk on a Free Group: Kesten Measure, Exact Excursions, and Radial Escape

HCS-C355 source-local reconstruction

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Abstract

For simple random walk on the rank-d free group, we give a single finite-to-infinite theorem joining operator spectrum, exact returns, and radial escape. With $D = 2d$, the Markov operator has purely absolutely continuous spectrum $[-2\sqrt{D-1}/D, 2\sqrt{D-1}/D]$ and root density $\sqrt{4(D-1) - D^2x^2}/[2\pi(1-x^2)]$. Weighted Dyck decomposition gives closed formulas for every even return and first return, including total return probability $1/(D-1)$. A pathwise boundary-flip coupling then proves speed $(D-2)/D$ and radial central-limit variance $4(D-1)/D^2$. The rank-one boundary becomes the recurrent walk on $\mathbb{Z}$, with arcsine spectrum and half-normal radial scaling. A 1,997-row exact certificate is a regression receipt, not a substitute for the all-time proofs. We claim neither priority nor an arithmetic-target interpretation.

Revision certificate. Kesten resolvent, root spectral law, radial chain, and exact even-return closure.

1 Frozen walk and complete theorem

Let $F_d = \langle a_1, \dots, a_d \rangle$, $d \geq 2$, and put $D = 2d$. Starting from the identity o, multiply on the right at each step by one element of $\{a_i, a_i^{-1} : 1 \leq i \leq d\}$, uniformly. Its Cayley graph is the D-regular tree T_D. On $\ell^2(T_D)$ the transition operator is the bounded self-adjoint operator

$$(Pf)(v) = D^{-1} \sum_{w \sim v} f(w).$$

Write $R_n = |X_n|$ for reduced-word length, and let $u_n = \mathbb{P}_o(X_n = o)$. Set $\rho = 2\sqrt{D-1}/D$ and $C_m = (m+1)^{-1}\binom{2m}{m}$.

Theorem 1 (Spectral, return, and escape atlas). *For the frozen walk:*

(i) *P is purely absolutely continuous and* $\operatorname{spec}(P) = [-\rho, \rho]$. *The spectral measure at o has density*

$$k_D(x) = \frac{\sqrt{4(D-1) - D^2x^2}}{2\pi(1-x^2)} 1_{\{|x|<\rho\}}. \tag{1}$$

Equivalently, for $z \notin [-\rho, \rho]$ and the branch $\sqrt{z^2 - \rho^2} \sim z$ at infinity,

$$\langle \delta_o, (z-P)^{-1}\delta_o \rangle = \frac{D\sqrt{z^2-\rho^2} - (D-2)z}{2(z^2-1)}. \tag{2}$$

(ii) *$R_0 = 0$, $R_{n+1} = 1$ from $R_n = 0$, while for $r \geq 1$,*

$$\mathbb{P}(R_{n+1} = r+1 \mid R_n = r) = p = \frac{D-1}{D}, \qquad \mathbb{P}(R_{n+1} = r-1 \mid R_n = r) = q = \frac{1}{D}. \tag{3}$$

All odd returns vanish and, for $n \geq 1$,

$$u_{2n} = D^{-2n} \sum_{k=1}^{n} \frac{k}{2n-k} \binom{2n-k}{n} D^k (D-1)^{n-k}. \tag{4}$$

(iii) *If $\tau_o^+ = \inf\{n \geq 1 : X_n = o\}$, then*

$$\mathbb{P}_o(\tau_o^+ = 2k) = \frac{C_{k-1}(D-1)^{k-1}}{D^{2k-1}}, \qquad \mathbb{P}_o(\tau_o^+ < \infty) = \frac{1}{D-1}. \tag{5}$$

(iv) *With $v = (D-2)/D$ and $\sigma^2 = 4(D-1)/D^2$,*

$$\frac{R_n}{n} \longrightarrow v \quad a.s., \qquad \frac{R_n - vn}{\sqrt{n}} \Longrightarrow N(0, \sigma^2). \tag{6}$$

(v) *At the excluded rank-one boundary $d = 1$, $F_1 \cong \mathbb{Z}$: P is purely absolutely continuous on $[-1,1]$ with arcsine density, $u_{2n} = 4^{-n}\binom{2n}{n}$, return is recurrent, $R_n/n \to 0$ a.s., and $R_n/\sqrt{n} \Rightarrow |N(0,1)|$.*

2 Cavity resolvent and the root measure

Delete the edge from a vertex to its parent and let $h(z)$ be the diagonal resolvent of the resulting rooted forward tree. Schur complementation at the root of that tree and then at o gives

$$h = \frac{1}{z - (D-1)h/D^2}, \qquad G_o(z) = \frac{1}{z - h/D}. \tag{7}$$

The solution satisfying $h(z) \sim z^{-1}$ is

$$h(z) = \frac{D^2}{2(D-1)}\big(z - \sqrt{z^2 - \rho^2}\big).$$

Substitution and rationalization prove (2). For $|x| < \rho$, the upper boundary value has

$$-\pi^{-1}\,\mathrm{Im}\, G_o(x + i0) = \frac{D\sqrt{\rho^2 - x^2}}{2\pi(1 - x^2)} = k_D(x).$$

The apparent singularities at $z = \pm 1$ in (2) are removable: the numerator vanishes there with the branch fixed above. There are therefore no exterior atoms. Stieltjes inversion gives (1) on the open band. Its integral is one (substitute $x = \rho\cos\theta$ and evaluate the resulting rational sine integral), while the square-root endpoints have no poles. Thus no singular-continuous or endpoint mass remains. The independent asymptotic check $zG_o(z) \to 1$ confirms the total normalization. Hence (1) is the full probability measure of the cyclic root vector in the radial subspace.

3 Weighted Dyck paths and exact returns

Distance from o changes by one. From a positive radius, precisely one letter cancels the last reduced letter and $D-1$ extend it; at radius zero all D letters lead to radius one. This proves (3). A return of length $2n$ is thus a Dyck path with n rises and n falls. Suppose it has exactly k irreducible excursions from zero. Its k rises from zero have D choices each; the remaining $n-k$ rises have $D-1$ choices, and every fall is forced.

It remains to count the paths. If $C(t) = 1 + tC(t)^2$ is the Catalan series, the required number is $[t^{n-k}]C(t)^k$. Lagrange inversion gives

$$[t^{n-k}]C(t)^k = \frac{k}{2n-k}\binom{2n-k}{n}. \tag{8}$$

Multiplying (8) by $D^k(D-1)^{n-k}$, summing over k, and dividing by D^{2n} proves (4). This also proves the renewal identity $u_{2n} = \sum_{k=1}^{n} \mathbb{P}(\tau_o^+ = 2k)u_{2n-2k}$ directly at word-count level.

References

[1] H. Kesten, “Symmetric random walks on groups,” *Trans. Amer. Math. Soc.* 92 (1959), 336–354. doi:10.1090/S0002-9947-1959-0109367-6; same record at doi:10.2307/1993160.

[2] W. Woess, *Random Walks on Infinite Graphs and Groups*, Cambridge Tracts in Mathematics 138, Cambridge University Press, 2000. doi:10.1017/CBO9780511470967.